

Homework 16.

1. $\nabla f(x) = \begin{pmatrix} 2x_1 + 4 \\ 4x_2 + 4 \end{pmatrix}$ $x^{k+1} = x^k + \alpha^k (-\nabla f(x^k))$

其中 $\alpha^k = \arg \min \{ f(x^k - \alpha \nabla f(x^k)) \mid \alpha \geq 0 \}$

α^k 满足 $\nabla f(x^k - \alpha^k \nabla f(x^k))^T \nabla f(x^k) = 0$

当 $k=0$ 时, $x^0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\nabla f(x^0) = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$, $\alpha \nabla f(x^0) = \begin{pmatrix} 4\alpha \\ 4\alpha \end{pmatrix}$

$\begin{pmatrix} -8\alpha + 4 \\ -16\alpha + 4 \end{pmatrix}^T \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 0 \Rightarrow -24\alpha + 8 = 0$
 $\alpha^1 = \frac{1}{3}$

$\Rightarrow x^1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} -\frac{4}{3} \\ -\frac{4}{3} \end{pmatrix}$

x^2 满足 $x^{k+1} = \begin{pmatrix} \frac{2}{3^k} - 2 \\ (\frac{1}{3})^k - 1 \end{pmatrix}$

利用数学归纳法完成证明.

2. $x^{k+1} \rightarrow \begin{pmatrix} -2 \\ -1 \end{pmatrix}$

伏格法计算

产销平衡

	B ₁	B ₂	B ₃	B ₄	B ₅	行差额				
						t ₂₁	t ₂₂	t ₂₃	t ₂₄	t ₂₅
A ₁	10	15	20	20	40	5	-	-	-	-
A ₂	20	40	15	30	30	5	5	10	10	-
A ₃	30	25	40	5	25	5	5	5	5	5
列差额	t ₁₁	10	20	5	10	5	-	-	-	-
	t ₁₂	10	5	-	25	5	-	-	-	-
	t ₁₃	10	5	-	15	5	-	-	-	-
	t ₁₄	10	5	-	-	5	-	-	-	-
	t ₁₅	30	25	-	-	25	-	-	-	-
	t ₁₆	30	25	-	-	25	-	-	-	-

销/产	B ₁	B ₂	B ₃	B ₄	B ₅	产量
A ₁		50				50
A ₂	10		60	30		100
A ₃	15	65			70	150
销量	25	115	60	30	70	